CISCO 3 IWS

DATA TRANSFER USING THE SONET PROTOCOL

Gadilshin Zhan & Nurbek Abildaev

Introduction to SONET

SONET (Synchronous Optical Networking) is a standard for transmitting digital data over optical fiber that synchronizes all network elements.

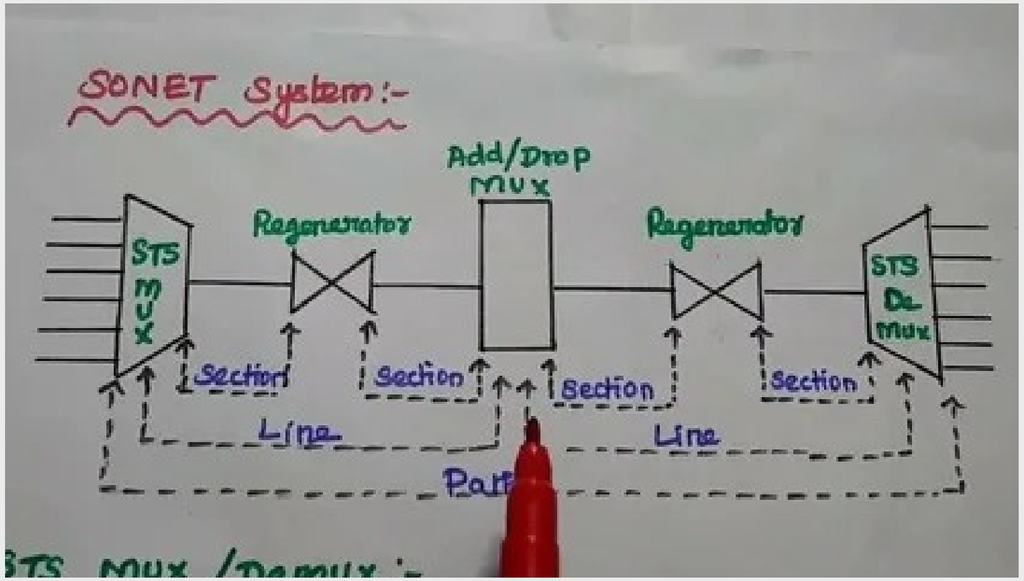
- ***Why it matters***: Before SONET, equipment from different vendors was often incompatible. SONET established a universal standard for signal formats and speeds.
- ***Core principle***: It is a synchronous network, meaning all nodes are clocked to a single, highly precise master reference source.

SONET Frame Structure

Dataistransmittedinfixed-sizeframesevery125 microseconds.

- Frame Components:
  - ***Overhead*** (Section/Line/Path): Management information for network control and error monitoring.
  - ***Payload***: The actual user data being transported.

- ***Advantage***: The fixed frame structure allows for Add-Drop Multiplexing (ADM)—extracting or inserting specific data channels without having to demultiplex the entire signal.



Speed Hierarchy (STS / OC)

SONET is highly scalable. The base unit is STS-1, and higher speeds are multiples of this base rate.

SDH	SONET	Скорость
	STS-1, OC-1	51,84 Мбит/с
STM-1	STS-3, OC-3	155,520 Мбит/с
STM-3	OC-9	466,560 Мбит/с
STM-4	OC-12	622,080 Мбит/с
STM-6	OC-18	933,120 Мбит/с
STM-8	OC-24	1,244 Гбит/с
STM-12	OC-36	1,866 Гбит/с
STM-16	OC-48	2,488 Гбит/с
STM-64	OC-192	9,953 Гбит/с
STM-256	OC-768	39,81 Гбит/с

STS: Synchronous Transport Signal (electrical level).
OC: Optical Carrier (converted into light for fiber transmission).

Network Topology & Self-Healing

SONETisfamousforitsringarchitecture,whichoffers extreme reliability.

Self-Healing: In the event of a fiber cut, the network automatically redirects traffic in the opposite direction.

Recovery time: The switch-over occurs in less than 50 milliseconds, making the failure virtually invisible to users.

Summary

Interoperability: Seamlesslyintegratesequipmentfromdifferentmanufacturers.
Reliability: Built-in faultdetectionandrapid"self-healing"restoration.
Manageability: Extensive overhead bytesallowforsophisticatednetwork monitoring.